Project Proposal - CPSC 4830

Page Limit: no more than 4 pages, but 3 pages are highly recommended.
PDF Format: NeurIPS template
(https://www.overleaf.com/latex/templates/neurips-2024/tpsbbrdqcmsh)

1. Research Problem Definition (10pts):
Define the problem that you will be investigating in this project. Consider addressing the following:

- Problem Statement: Clearly define the specific problem.
- Motivation & Significance: Why is this problem interesting or socially/technically important?
- Learning Objective: What exactly is the deep learning model tasked to "learn" (e.g., node labels, edge existence, graph-level properties)?
- Success Criteria: What would a "successful" project look like by the end of the semester?

2. Dataset (20pts):
Which dataset will you use? Provide a detailed description of the dataset you plan to use for this project:

- Description: Source of the data and its real-world meaning.
- Exploratory Data Analysis (EDA), such as:
  - Graph Statistics: Number of nodes/edges, density, average clustering coefficient, and degree distribution.
  - Feature Analysis: Brief description of node/edge attributes and label distribution (check for class imbalance).
- Visualizations: If possible, provide at least one meaningful visualization (e.g., a sample subgraph or a t-SNE plot of initial features).

3. Description of Related Works (20pts):
Conduct a comprehensive literature review regarding your project:

- Breadth & Depth: Cite at least 3–5 high-quality papers relevant to the domain or methodology.
  - If you want to propose a new GNN algorithm, review similar algorithms; if your goal is to apply an algorithm to a new domain, examine how similar approaches have been adapted in other domains
- Comparative Analysis: Do not just list papers. Explain how these works succeed, where they fail, and exactly how your project fills a gap or explores a new variation.
- Positioning: Clearly state: "Unlike [Author et al.], our project focuses on..."

4. Proposed Approaches (30pts):

- Individual Contribution: Each member must have a dedicated subsection (e.g., Section 4.1, 4.2).
  - Each team member (if applicable) should implement their own proposed approach. While methods may have shared elements, they should have unique elements.
- Technical Justification: Why are these specific architectures (e.g., GCN, GAT, GraphSAGE) chosen for your dataset?
- Adaption/Adjustment: If using existing code, you must specify your non-trivial modification, or combine two repositories for an application.
  - *Examples:* Implementing a custom loss function, adding an attention mechanism to a standard model, or a feature engineering pipeline.
- Research Questions and Hypothesis: Briefly describe your research questions and hypotheses. These questions should be more detailed than the general research questions outlined in item 1.
  - For instance, if your overall research question is to propose a new clustering algorithm, your specific research questions could be "what's the performance difference between my clustering algorithm and heuristic clustering algorithms, such as k-means, dbscan, and mean shift?" Your hypothesis could be "my proposed algorithm will run faster than any of these heuristics, while maintaining high accuracy."

5. Evaluation Metrics and Experiment Setup (10pts):

- Baselines: List the "standard" models you will compare your results against (e.g., a simple MLP, a vanilla GCN, or a heuristic).
- Metrics: Choose appropriate metrics (Accuracy, F1-score, AUROC, or RMSE). Justify why these metrics matter for your specific problem (especially if the data is imbalanced).
- Validation Strategy: Mention if you will use a fixed split or K-fold cross-validation.

6. Timeline (10pts):

- Milestones: Break the project into phases (Data Preprocessing, Model Implementation, Hyperparameter Tuning, Final Analysis).
- Responsibility Matrix: A simple table showing who is responsible for which task to ensure accountability.